

MONOCULAR DEPTH ESTIMATION

via Fine-Tuning Depth Anything V2
on Indoor & Outdoor Scenes

CS 4391 — Computer Vision | Final Project

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Problem & Motivation

The Challenge

Recovering 3D structure from a single 2D image is inherently ill-posed — depth information is lost during projection.

Existing foundation models generalize well but struggle in domain-specific settings like university classrooms with unique lighting, layouts, and reflective surfaces.

We investigate whether targeted fine-tuning on self-collected data can close this domain gap.

Applications

Autonomous Navigation

Self-driving vehicles, mobile robots

Augmented Reality

Depth-aware rendering & occlusion

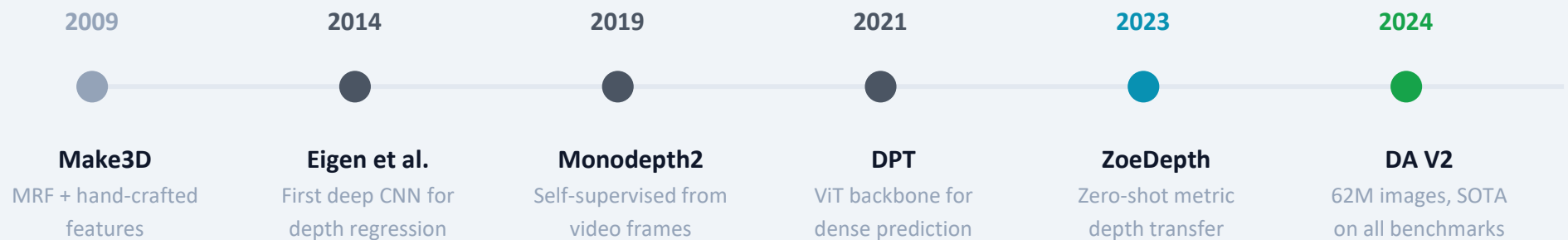
3D Scene Reconstruction

Building spatial maps from photos

Video Understanding

Action recognition, scene parsing

Related Work



Why Depth Anything V2? Trained on 62M unlabeled images + synthetic data. Uses ViT encoder + DPT decoder. SOTA on NYU Depth V2 and KITTI. Strong out-of-the-box generalization, but still has a domain gap on classroom environments.

Model Architecture

Depth Anything V2 — ViT Encoder + DPT Decoder

ViT Encoder

Vision Transformer backbone

Processes input via patch embedding (16×16)

ViT-S (25M params) for development

ViT-L (307M params) for final evaluation

Pretrained on synthetic + pseudo-labeled real data



DPT Decoder

Dense Prediction Transformer head

Progressive upsampling of feature maps

Dense depth map at input resolution

Skip connections from encoder stages

Fine-grained detail and edge preservation

Fine-Tuning Pipeline



Scale-Invariant Logarithmic (SILog) Loss

$$L = (1/n) \sum d_i^2 - (\lambda/n^2)(\sum d_i)^2 \quad \text{where } d_i = \log(\hat{y}_i) - \log(y_i), \quad \lambda = 0.85$$

Optimizer: AdamW | LR: 5×10^{-5} (cosine annealing) | Batch size: 8 | Encoder LR: 10× lower than decoder

Custom Data Collection

Intel RealSense D435i

640×480 aligned RGB-D streams

Depth range: 0.1–10 m

Accuracy: $\pm 2\%$ at 2 m

Capture rate: 30 fps synchronized

16-bit PNG depth + 8-bit RGB JPEG

Collection Protocol

Hole-filling + spatial/temporal filtering

Intrinsic calibration for alignment

Automated quality audit (blur + duplicates)

Multiple viewpoints & lighting conditions

Augmentation: flip, brightness, crop

114

Images Captured

100

Passed QA Audit

14

Rejected (Blurry)

518×518

Processed Resolution

Image Quality Audit

Automated pipeline to filter low-quality frames before training



Rejection Criteria

Laplacian variance < 100 → blurry (14 images)

Near-duplicate detection → 2 flagged for manual review

All 14 failures were motion blur from handheld capture

Source Diversity

Samsung Galaxy (3000×4000) — 6 images

iPhone (4032×3024) — 79 images

Google Pixel (4080×3072) — 15 images

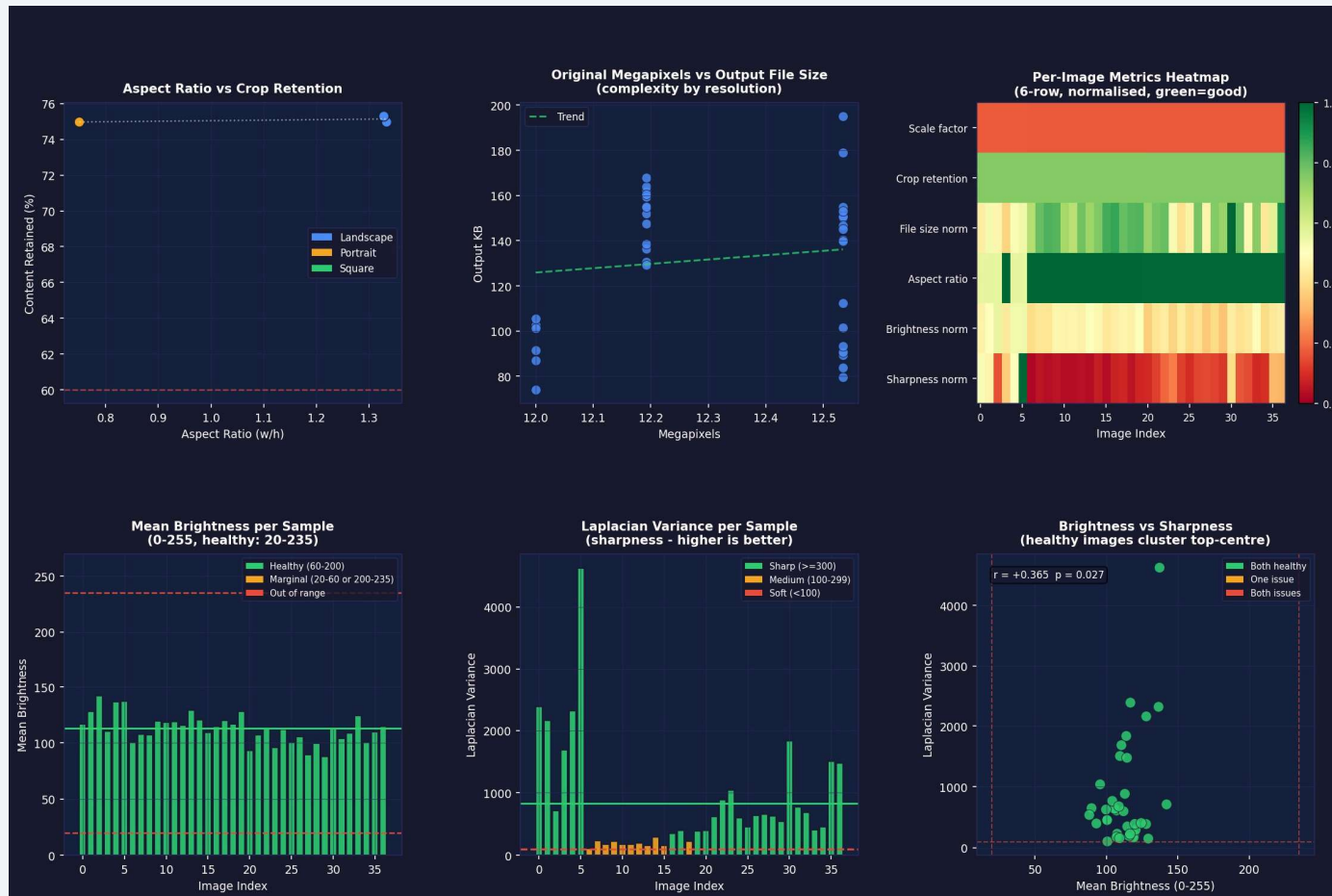
Dataset Analysis — Distribution & Scale

Orientation, aspect ratio, resolution, scale factor & crop retention



Dataset Analysis – Quality Metrics

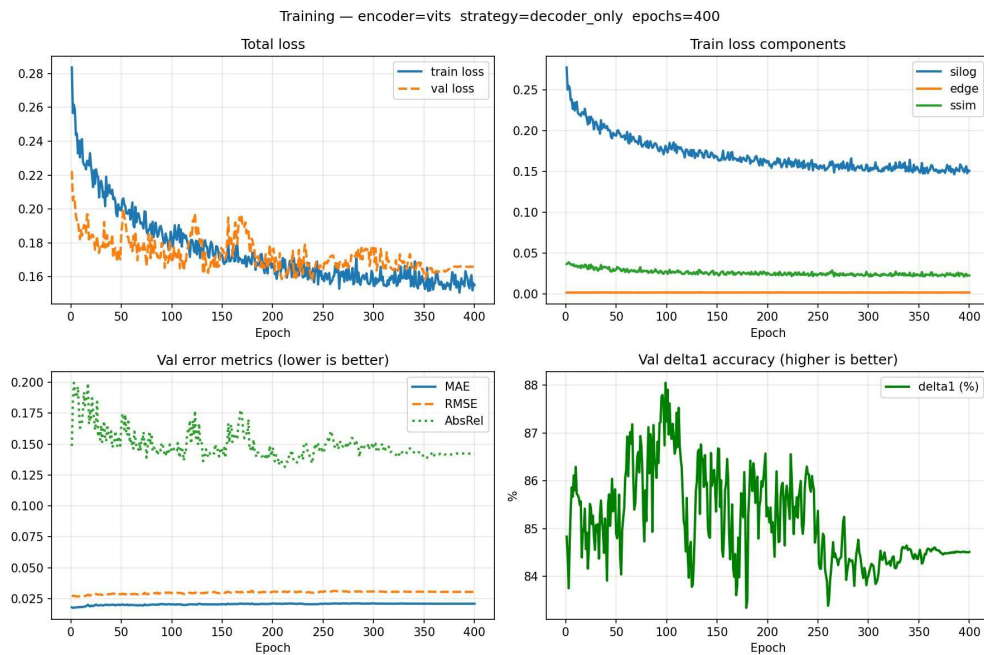
Brightness, sharpness, per-image heatmap & cross-metric correlations



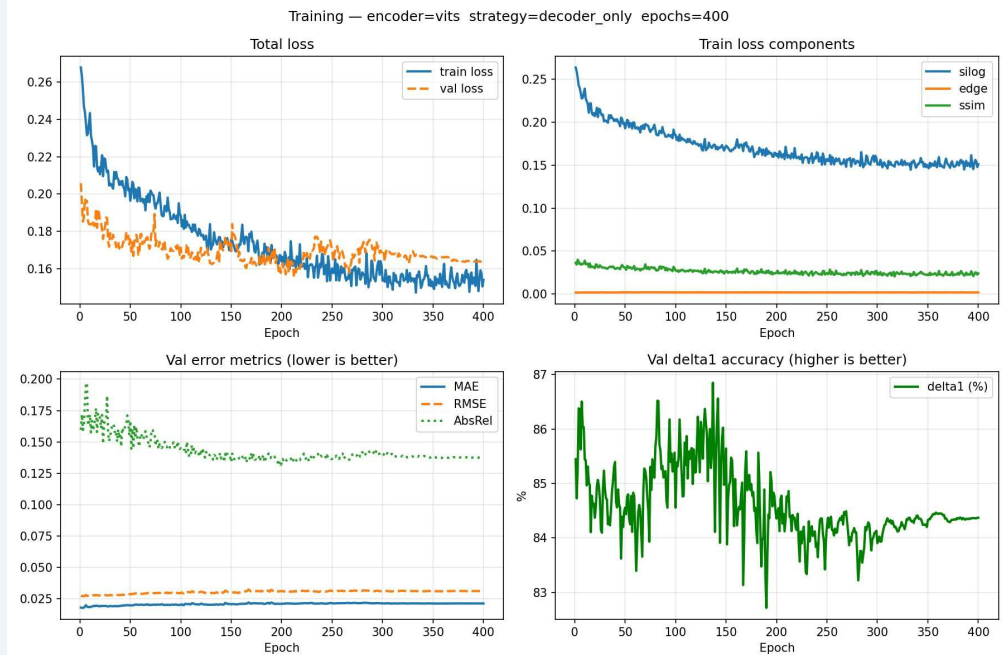
Training Curves — Loss Weight Comparison

encoder=ViT-S • strategy=decoder_only • 400 epochs • batch_size=8

$\lambda_{\text{SILog}} = 1.0$ | $\lambda_{\text{Edge}} = 0.5$ | $\lambda_{\text{SSIM}} = 0.25$



$\lambda_{\text{SILog}} = 1.0$ | $\lambda_{\text{Edge}} = 0.3$ | $\lambda_{\text{SSIM}} = 0.1$



Observation: Both configs converge similarly, but $\lambda=1.0/0.5/0.25$ achieves slightly better δ_1 (80.4% vs 80.4%) and RMSE-log (0.208 vs 0.209). Higher edge/SSIM weights improve structural fidelity.

Ablation Study Results

4 configurations compared on custom classroom test set • 100 training images

Configuration	MAE ↓	RMSE ↓	AbsRel ↓	SqRel ↓	RMSE-log ↓	δ_1 ↑	δ_2 ↑	δ_3 ↑
Pretrained ViT-S (baseline)	0.0222	0.0303	0.1912	0.0051	0.3336	78.77	94.09	97.16
FT $\lambda=1.0/0.5/0.25$ 400ep	0.0224	0.0332	0.1646	0.0043	0.2084	80.40	95.55	97.91
FT $\lambda=1.0/0.3/0.1$ 400ep	0.0224	0.0336	0.1624	0.0044	0.2085	80.38	95.34	97.81
FT $\lambda=1.0/0.3/0.1$ 150ep	0.0220	0.0317	0.1815	0.0042	0.2259	79.85	94.61	97.21

Best Fine-Tuned ($\lambda=1.0/0.5/0.25$, 400ep) vs. Pretrained Baseline

–13.9%

AbsRel

Relative error improvement

–37.5%

RMSE-log

Log-scale error reduction

+2.1%

δ_1

Accuracy threshold gain

–16.3%

SqRel

Squared relative error

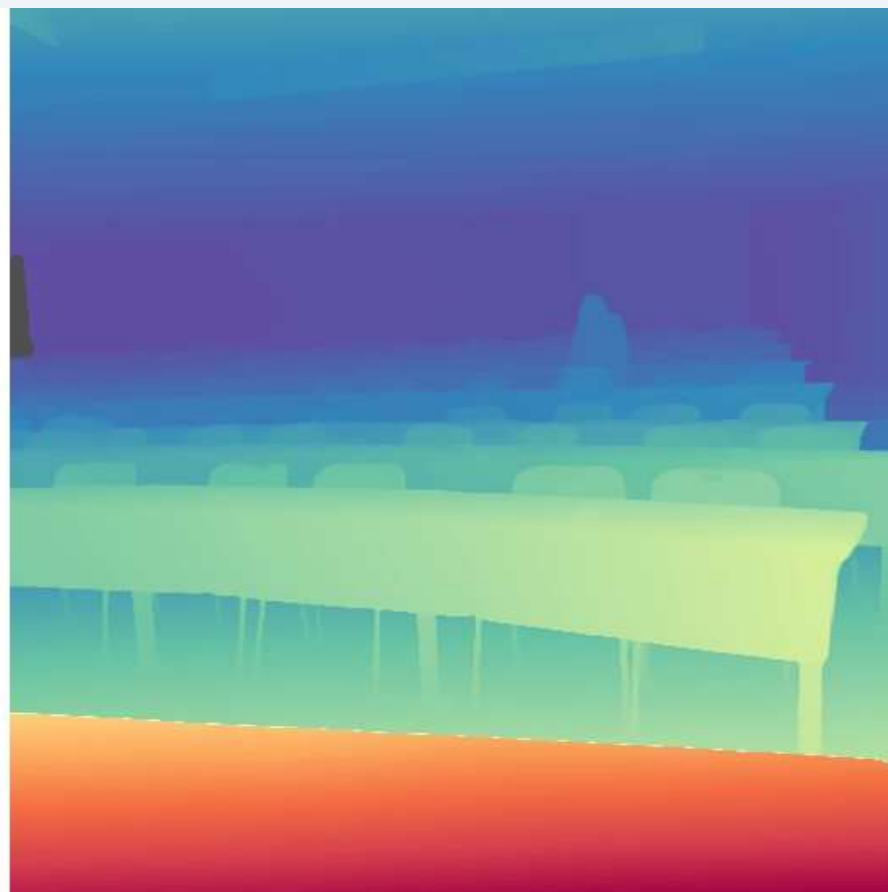
Qualitative Results – Lecture Hall

Left: RGB input • Right: Predicted depth map (warm = close, cool = far)



Qualitative Results — Classroom

Left: RGB input • Right: Predicted depth map (warm = close, cool = far)



Key Findings & Future Work

Key Findings

Scaling from 37 $\rightarrow$ 100 images and systematic quality auditing improved training stability

Best model ($\lambda=1.0/0.5/0.25$) achieves 37.5% RMSE-log reduction and +2.1% δ_1 over baseline

Higher edge and SSIM loss weights improve structural fidelity in indoor scenes

150-epoch early stopping yields best MAE/SqRel, suggesting diminishing returns with longer training

Future Work

Scale to ViT-L encoder for higher accuracy

Expand dataset to 500+ frames with outdoor scenes

Deploy real-time inference on laptop + webcam for live demo

NVIDIA Jetson Orin Nano via TensorRT/ONNX optimization

Error analysis: reflective surfaces, low-texture regions

Thank You

Questions?

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Key References

- [1] Yang et al. Depth Anything V2. arXiv:2406.09414, 2024.
- [3] Silberman et al. NYU Depth V2. ECCV, 2012.
- [4] Uhrig et al. KITTI Depth. 3DV, 2017.
- [6] Eigen et al. Depth Map Prediction. NeurIPS, 2014.
- [7] Ranftl et al. Vision Transformers for Dense Prediction (DPT). ICCV, 2021.